

Bluestock Mutual Fund Capstone Project

Final Report

Data Engineering & Analytics Capstone
Indian Mutual Fund Industry

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1 Executive Summary

This report presents the complete findings of the Bluestock Mutual Fund Capstone Project - a multi-phase data engineering and analytics initiative covering the Indian mutual fund industry. The project ingested, cleaned, and warehoused 10 datasets comprising over 100,000 records, built a star-schema SQLite database, performed comprehensive exploratory and performance analytics, and delivered an interactive Tableau dashboard.

Key Outcomes

- 40 mutual fund schemes across 10 fund houses analysed end-to-end.
- 64,320 daily NAV rows after calendar forward-fill (from 46,000 raw rows).
- SIP inflows reached Rs 31,002 Cr all-time high in December 2025.
- Industry folios grew from 13.26 Cr to 26.12 Cr (nearly doubled).
- Composite fund scorecard built with 5-factor scoring (CAGR, Sharpe, Alpha, Expense, Drawdown).
- VaR/CVaR computed for all 40 schemes; rolling 90-day Sharpe tracked.
- 4-page interactive Tableau dashboard delivered with 17 visualisations.

Project Phases

Day	Phase	Key Deliverables
1	Data Ingestion	10 CSVs loaded, profiled, AMFI validated
2	Cleaning + SQLite	Star schema, 11 tables, quality reports
3	EDA	18 charts, Jupyter notebook, 10 findings
4	Performance	Scorecard, CAGR, Sharpe, alpha/beta
5	Dashboard	4-page Tableau workbook
6	Advanced Analytics	VaR, HHI, cohort, SIP continuity

2 Data Sources

The project uses 10 structured CSV datasets representing different facets of the Indian mutual fund industry. An additional live NAV feed from mfapi.in supplements historical data for 6 bluechip schemes.

Raw Dataset Inventory

#	Dataset	Records	Key Columns
01	Fund Master	40	amfi_code, fund_house, category, risk
02	NAV History	46,000	amfi_code, date, nav
03	AUM by Fund House	~80	fund_house, aum_crore, date
04	Monthly SIP Inflows	48	month, sip_inflow_crore
05	Category Inflows	~240	month, category, net_inflow_crore
06	Industry Folio Count	12	month, total_folios_crore
07	Scheme Performance	40	returns, alpha, beta, sharpe
08	Investor Transactions	50,000	investor_id, amount, state, age
09	Portfolio Holdings	~400	stock, sector, weight_pct
10	Benchmark Indices	~4,000	index_name, date, close_value

Live Data Feed

The live_nav_fetch.py script queries the MFAPI (<https://api.mfapi.in>) for historical NAV data for 6 flagship schemes: HDFC Top 100, SBI Bluechip, ICICI Bluechip, Nippon Large Cap, Axis Bluechip, and Kotak Bluechip.

Data Quality Profile

Day 1 profiling revealed zero critical data issues across all 10 files. All AMFI codes in fund_master were validated against nav_history with zero missing matches. Day 2 cleaning expanded 46,000 raw NAV rows to 64,320 after calendar gap filling, excluded 0 invalid transactions, and flagged 0 scheme performance anomalies.

3 ETL Design

Architecture Overview

The ETL pipeline follows a classic Extract -> Transform -> Load pattern, implemented in Python with Pandas for transformation and SQLAlchemy for loading into a SQLite star-schema warehouse.

Pipeline flow: Raw CSVs -> Python cleaning scripts -> Processed CSVs -> SQLite star schema -> Analytics / Tableau

Star Schema Design

The SQLite warehouse uses a star schema with 2 dimension tables and 4 fact tables, plus 5 auxiliary tables for industry-level metrics.

Table	Type	Grain	Row Count
dim_fund	Dimension	One row per AMFI scheme	40
dim_date	Dimension	One row per calendar date	~1,500
fact_nav	Fact	Scheme x calendar date	64,320
fact_transactions	Fact	One row per transaction	50,000
fact_performance	Fact	One row per scheme	40
fact_aum	Fact	Fund house x date	~80

Key Cleaning Rules

- NAV History: parsed dates, sorted, deduplicated, forward-filled calendar gaps, validated NAV > 0.
- Transactions: standardised types (SIP/Lumpsum/Redemption), validated amounts > 0, KYC enum check.
- Performance: coerced numeric fields, flagged returns outside +/-100%, expense ratio validated 0.1-2.5%.
- All tables: duplicate removal, date parsing, foreign-key integrity via date_key joins.

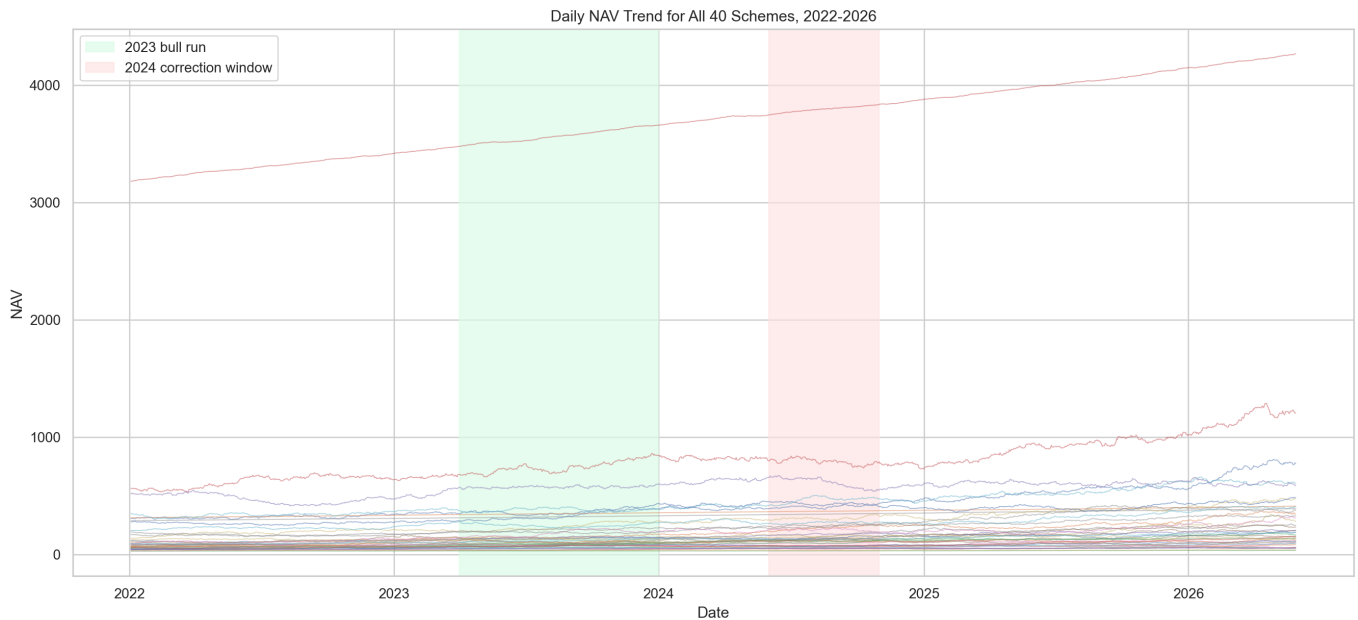
Calendar Forward-Fill

Missing calendar dates within each scheme's NAV history were filled using forward-fill (ffill). This expanded the dataset from 46,000 raw rows to 64,320 daily rows, with 18,320 synthetic rows inserted. This ensures continuous time-series for return calculations.

4 EDA Findings

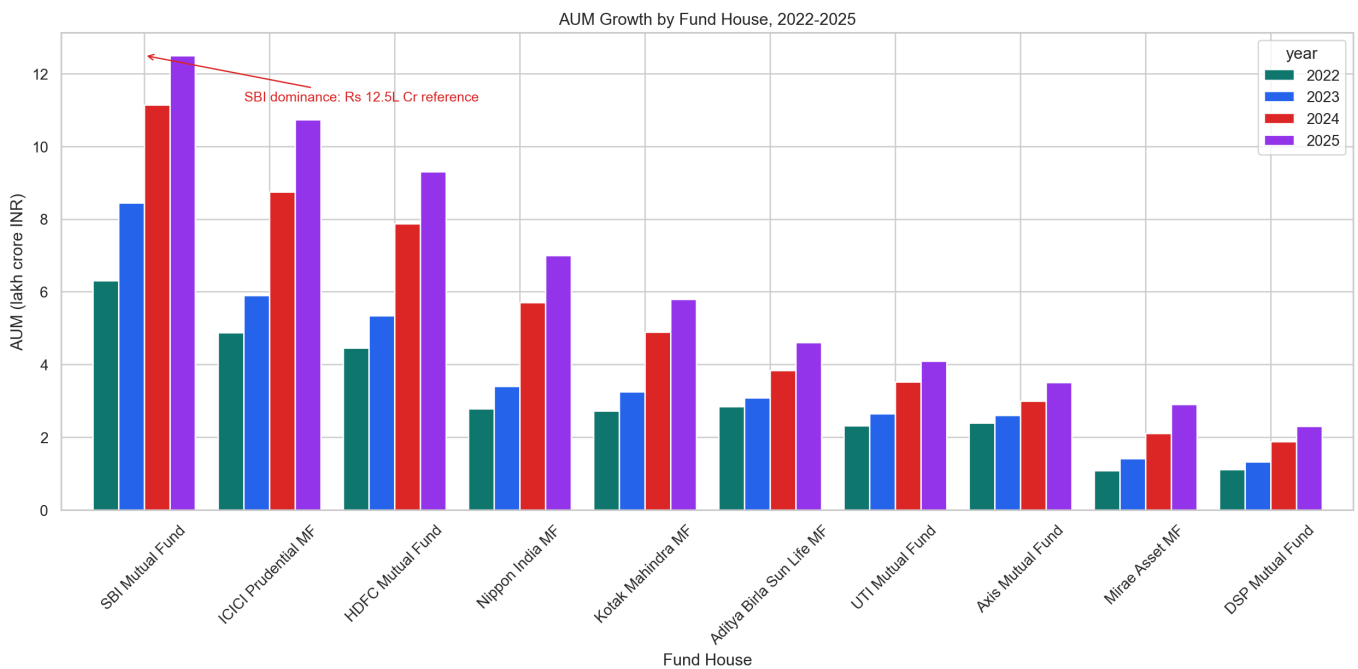
Day 3 exploratory analysis generated 18 publication-quality charts examining NAV trends, AUM concentration, SIP inflows, investor demographics, portfolio composition, and risk-return profiles.

4.1 NAV Trends



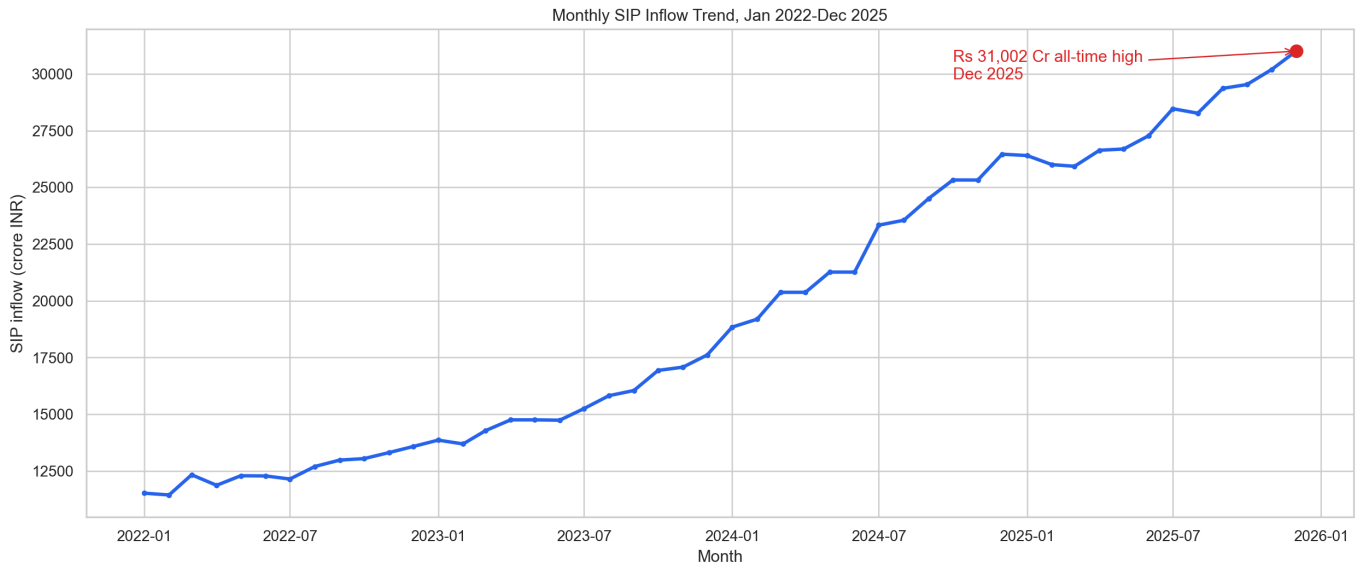
Daily NAV trends for all 40 schemes (2022-2026) with the 2023 bull-run and 2024 correction windows highlighted. NAV levels vary significantly across schemes.

4.2 AUM Growth



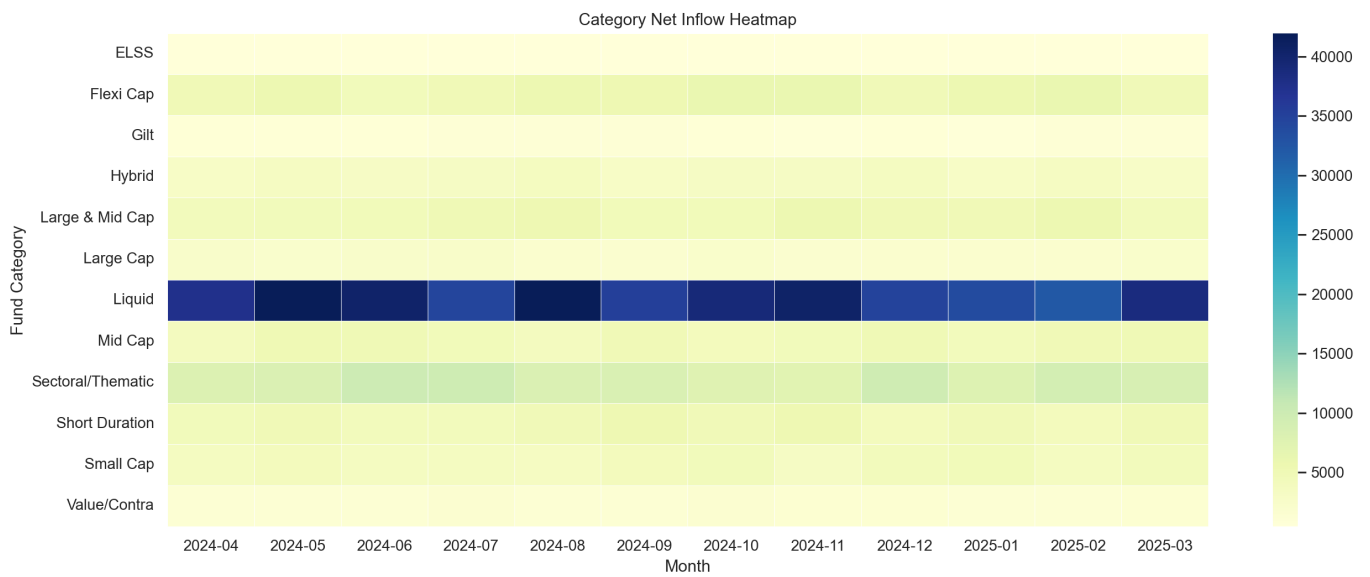
SBI Mutual Fund leads as the dominant AUM leader.

4.3 SIP Inflows

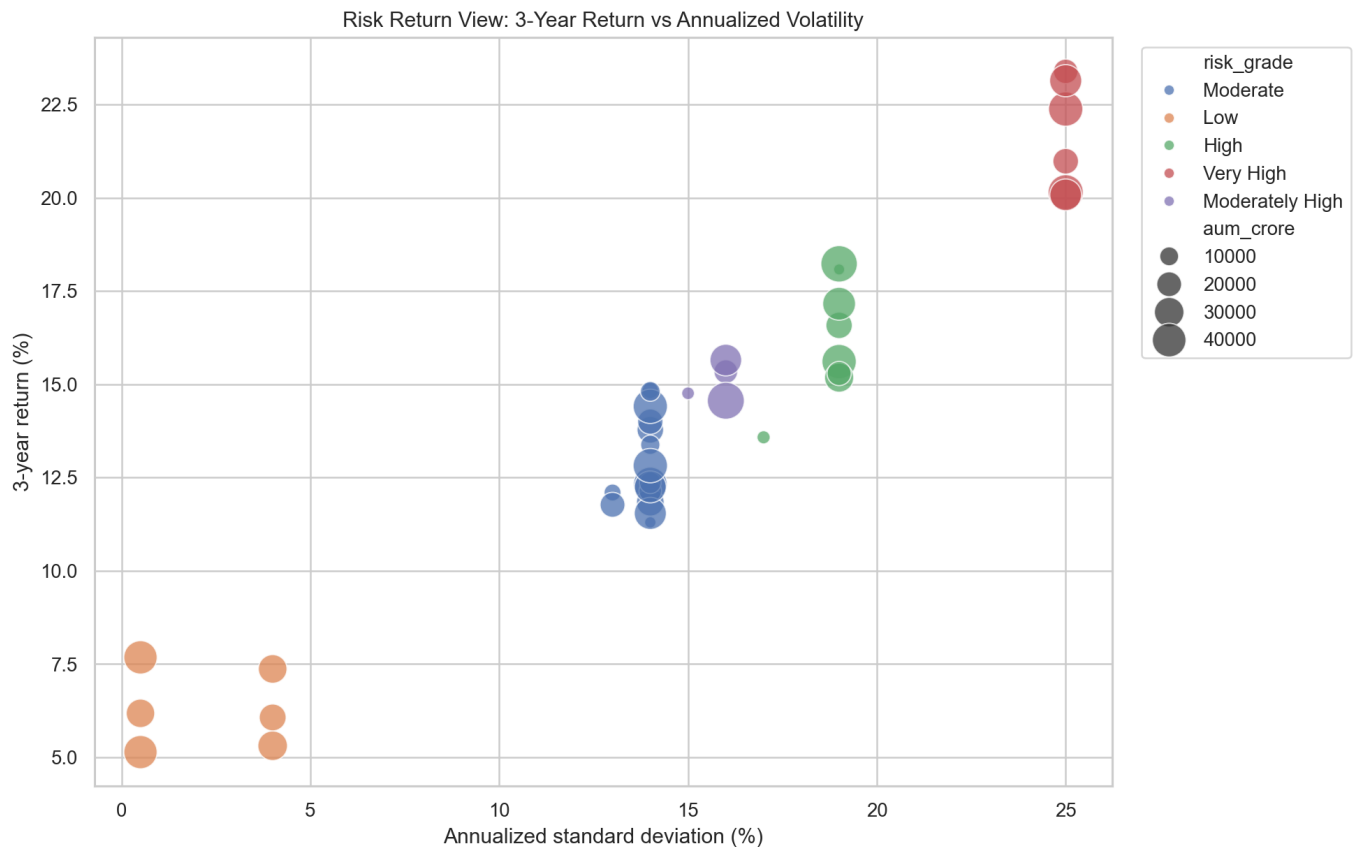


SIP inflows grew steadily and reached an all-time high of Rs 31,002 Cr in Dec 2025.

4.4 Category Inflows



4.5 Risk-Return Profile



Risk-return scatter shows the trade-off between 3-year returns and annualised volatility, with bubble size proportional to AUM.

10 Key EDA Findings

- NAV levels vary widely; indexed views are required before comparing growth.
- Several schemes compounded meaningfully after the 2023 rally window.
- SBI Mutual Fund is the clear AUM leader with Rs 12.5L Cr dominance.
- SIP inflows reached Rs 31,002 Cr all-time high in Dec 2025.
- Category inflows are uneven, with concentrated bursts in selected categories.
- Investor participation is distributed across multiple age bands.
- SIP ticket sizes differ by age group with notable outliers.
- SIP contribution is geographically concentrated in top states.
- Industry folios approximately doubled from 13.26 Cr to 26.12 Cr.
- Selected fund returns are positively correlated, limiting diversification benefit.

5 Performance Analysis

5.1 Scorecard Methodology

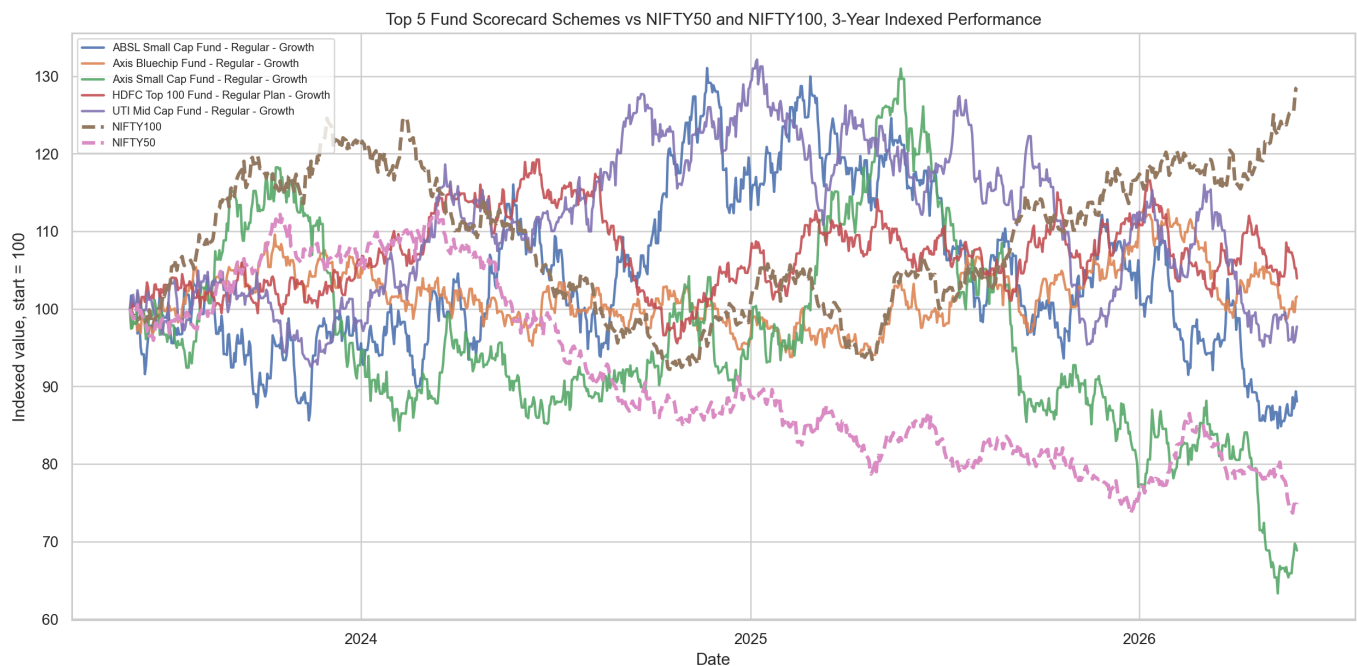
A composite 0-100 fund score was built from five percentile-ranked factors:

Factor	Weight	Direction
3-Year CAGR	30%	Higher is better
Sharpe Ratio	25%	Higher is better
Alpha (vs NIFTY100)	20%	Higher is better
Expense Ratio	15%	Lower is better
Max Drawdown	10%	Less negative is better

5.2 Risk Metrics

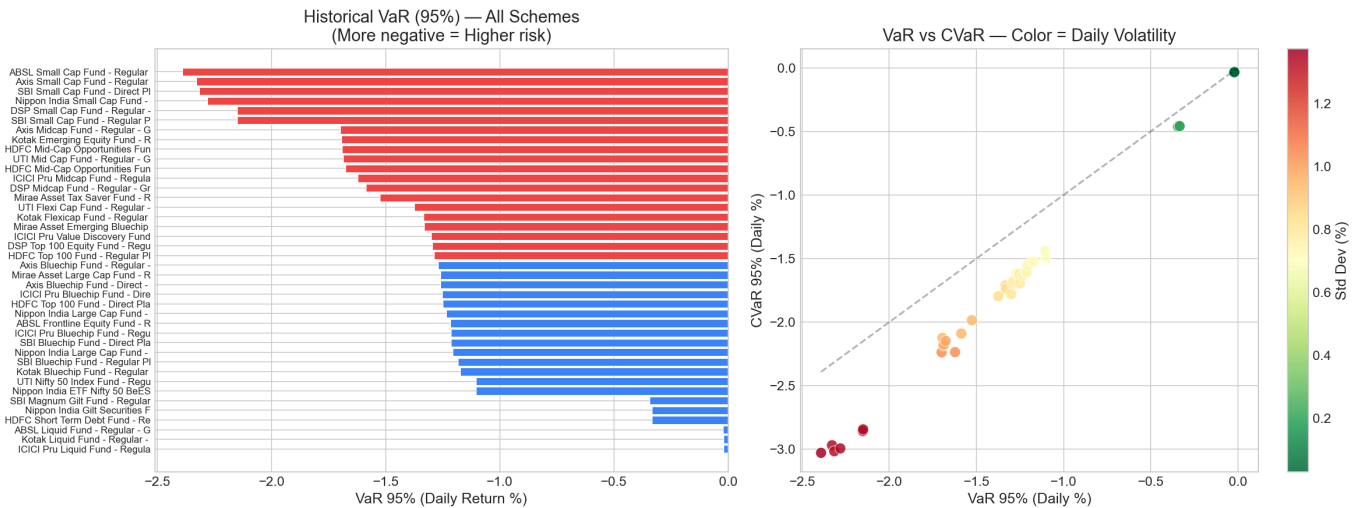
Daily returns were computed as $\text{NAV}(t)/\text{NAV}(t-1) - 1$ for all 40 schemes. Sharpe Ratio uses a 6.5% annual risk-free rate proxy. Sortino Ratio uses downside volatility only. Alpha and beta are estimated via OLS regression against NIFTY100.

5.3 Benchmark Comparison



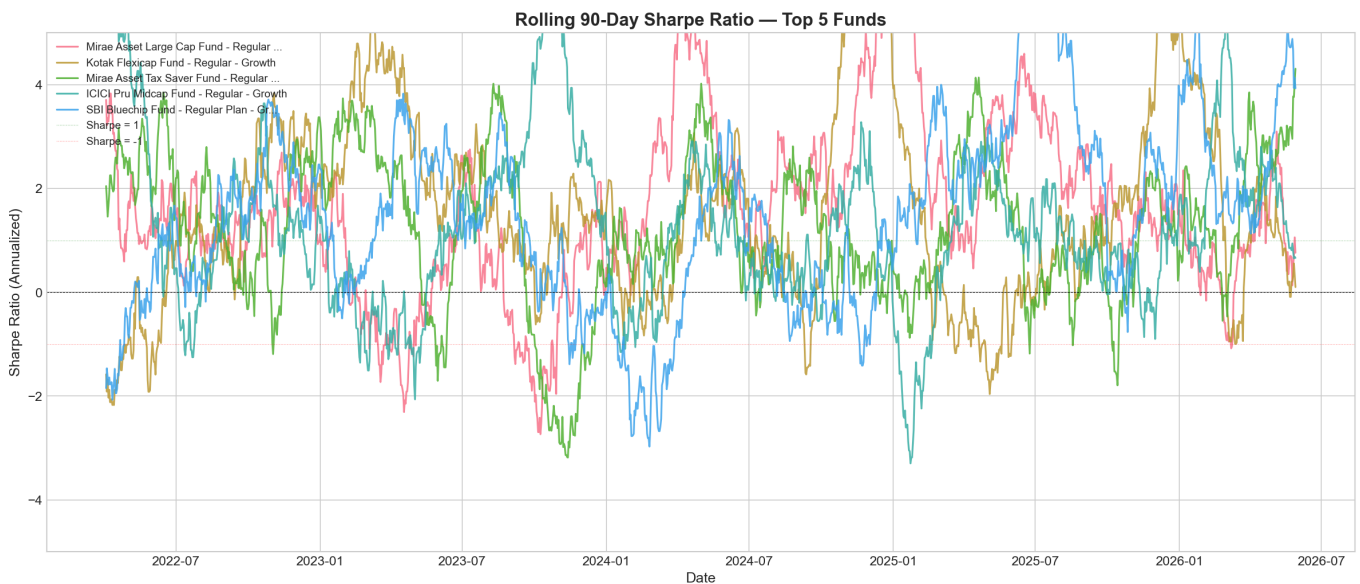
3-year indexed performance of the top 5 scorecard funds vs NIFTY50 and NIFTY100 benchmarks.

5.4 Value-at-Risk Analysis



Historical VaR (95%) and CVaR were computed for all 40 schemes. VaR represents the worst daily loss expected on 19 out of 20 days. CVaR (Expected Shortfall) is the average loss on the worst 5% of days.

5.5 Rolling Sharpe Ratio



90-day rolling Sharpe Ratio for the top 5 funds by overall Sharpe, showing how risk-adjusted performance evolves over time.

6 Dashboard

A 4-page interactive Tableau dashboard was built to visualise all analytical outputs. Below are screenshots of each dashboard page.

Industry Overview: KPIs, AUM timeline, fund house ranking

1. Industry Overview

P1_KPI_T

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Rs. 81.00L Cr

P1_KPI_S

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Rs. 31,002 Cr

P1_KPI
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Rs. 26.12 Cr

P1_K

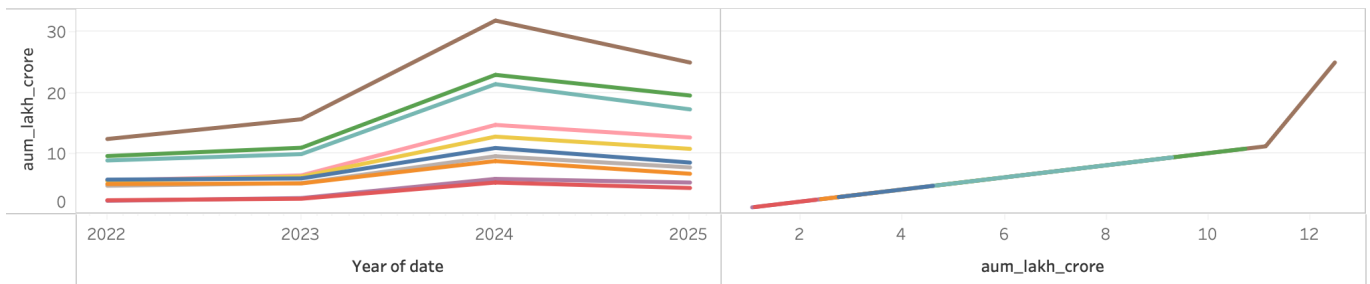
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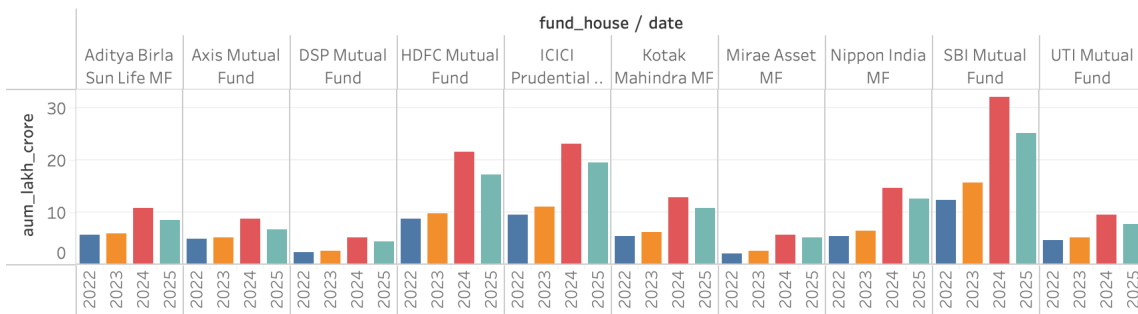
1,908



P1_Industry_AUM_Timeline



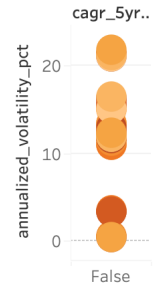
P1_AUM_by_Fund_House



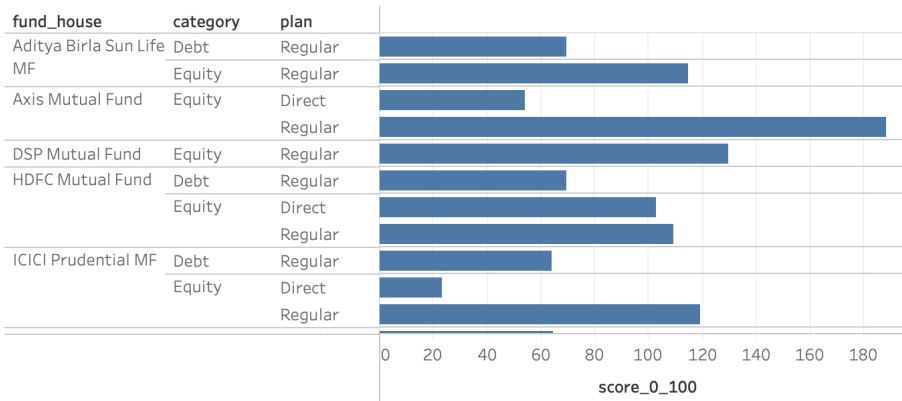
Fund Performance: Risk-return scatter, scorecard, NAV comparison

2. Fund Performance

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P2_Fund_Scorecard_Table

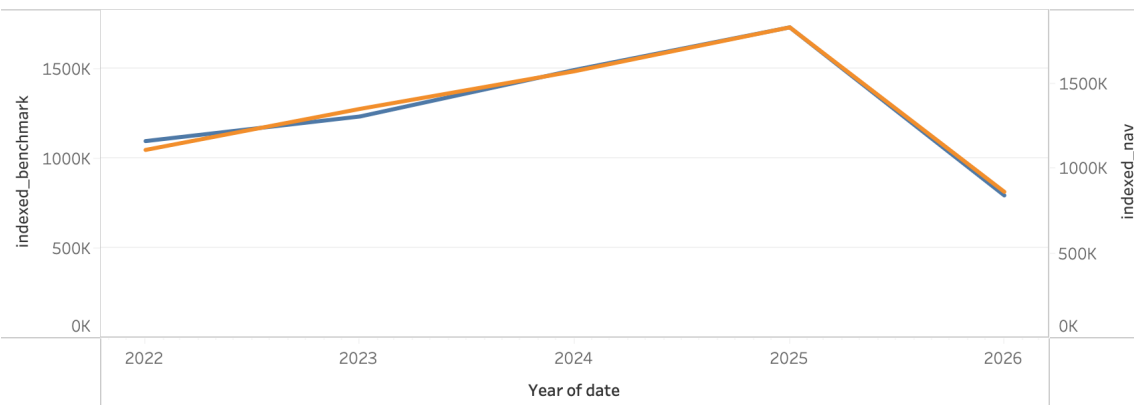


category
☒ Debt
☒ Equity

plan
☒ Direct
☒ Regular

fund_house
☒ Aditya Birla Sun Life ..
☒ Axis Mutual Fund
☒ DSP Mutual Fund
☒ HDFC Mutual Fund
☒ ICICI Prudential MF
☒ Kotak Mahindra MF
☒ Mirae Asset MF
☒ Nippon India MF
☒ SBI Mutual Fund
☒ UTI Mutual Fund

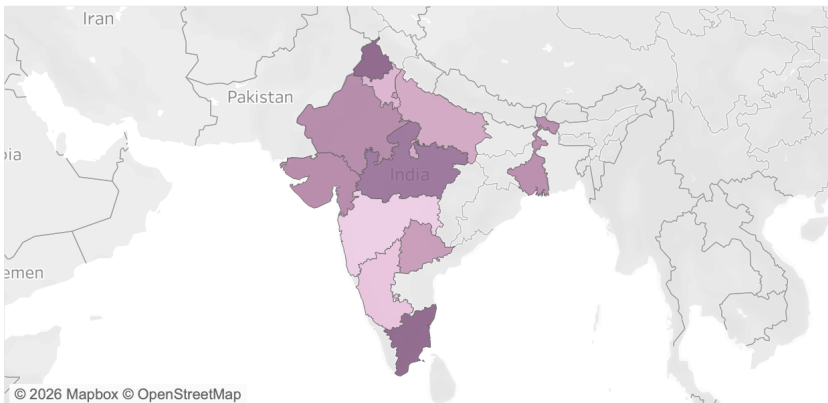
P2_Fund_NAV_vs_Benchmark



Investor Analytics: Demographics, geography, transaction patterns

3. Investor Analytics

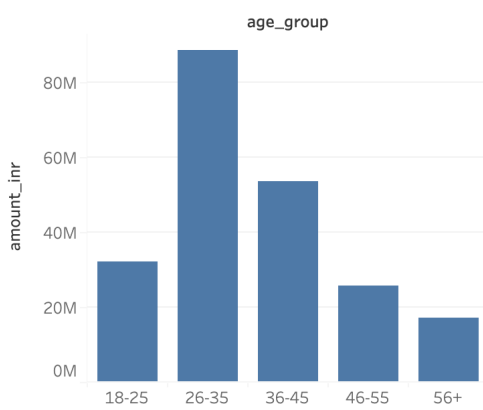
P3_Transactions_by_State



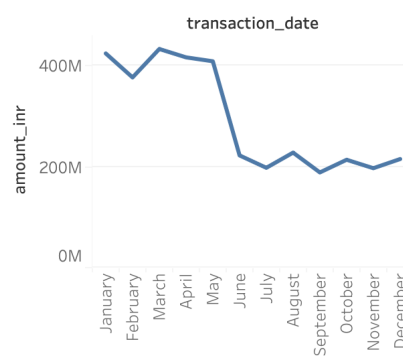
P3_Transaction_Type_Donut

Tableau can't export Viz Extensions.

P3_SIP_by_Age_Group

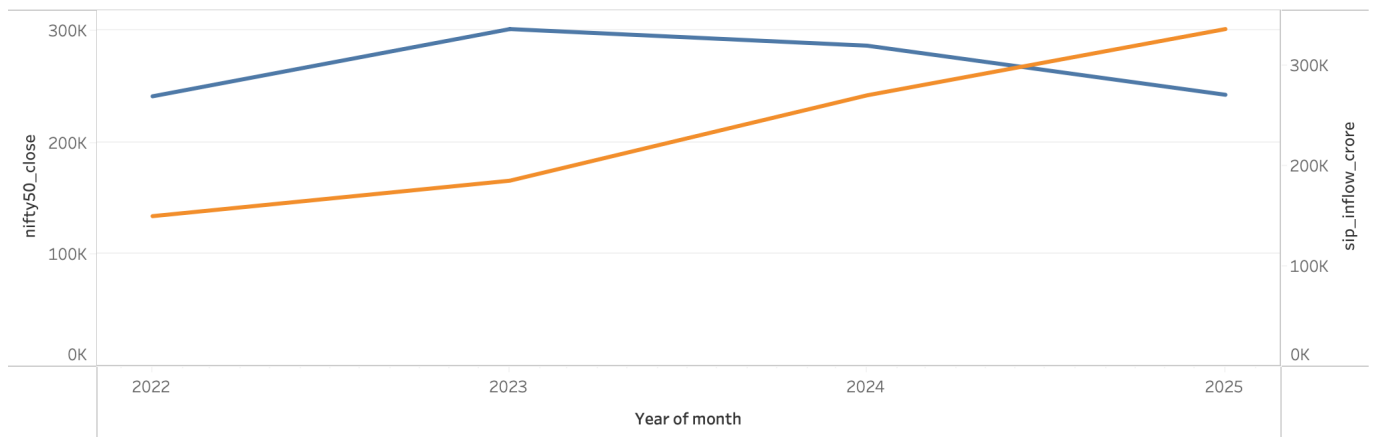


P3_Monthly_Transaction_Volume

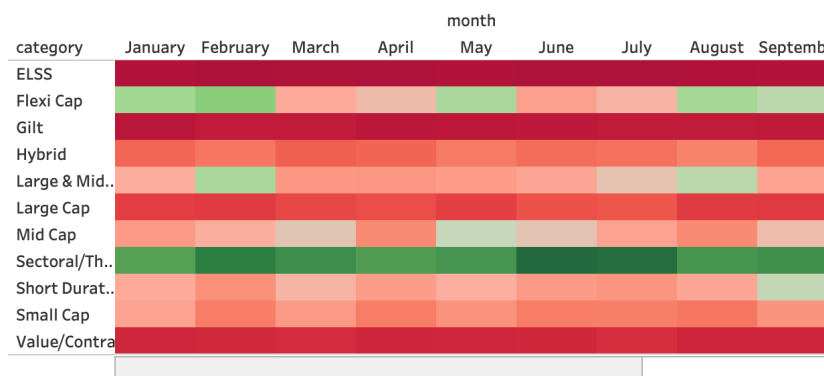


SIP & Market Trends: SIP vs Nifty, category inflows, YoY growth

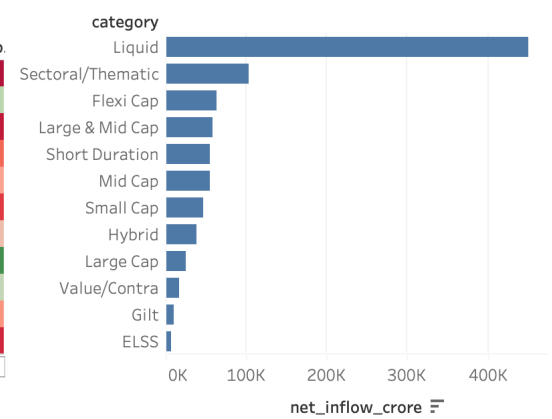
P4_SIP_Inflow_vs_Nifty



P4_Category_Inflow_Heatmap



P4_Top_Categories_FY25



7 Limitations

5-Year CAGR Unavailable

The cleaned NAV history starts in January 2022, providing only ~4 years of data. True 5-year CAGR cannot be computed and is flagged as unavailable in the scorecard.

Synthetic Investor Data

The investor transactions dataset is synthetically generated for capstone purposes. Demographic distributions may not reflect real-world patterns.

No Real-Time Pipeline

The ETL pipeline runs as a batch process. There is no streaming or scheduled ingestion. NAV data requires manual refresh via `live_nav_fetch.py`.

Forward-Fill Bias

Calendar gaps in NAV history are forward-filled, which creates zero-return days. This slightly dampens volatility estimates and may understate true risk.

Single Benchmark Regression

Alpha and beta are computed against NIFTY100 only. Multi-factor models (Fama-French) would provide more robust risk attribution.

Static Dashboard

The Tableau workbook uses flat CSV extracts. It does not connect to a live database and requires manual data refresh.

8 Recommendations

Real-Time API Integration

Connect the pipeline to MFAPI or AMFI feeds for automated daily NAV ingestion using Apache Airflow or a scheduled cron job.

ML-Based Fund Recommender

Replace the rule-based recommender with a collaborative filtering or content-based model trained on investor transaction patterns and fund features.

Multi-Factor Risk Model

Extend alpha/beta analysis with Fama-French 3-factor or Carhart 4-factor models for more robust risk-adjusted performance attribution.

SIP Churn Prediction

Build a classification model to predict SIP discontinuation using the SIP continuity features (gap days, frequency, amount trends) as input features.

Portfolio Optimisation

Implement mean-variance (Markowitz) or risk-parity portfolio optimisation to suggest optimal fund allocations based on investor risk profiles.

Cloud Deployment

Migrate the SQLite warehouse to PostgreSQL on a cloud platform (AWS RDS / GCP Cloud SQL) and deploy the dashboard via Tableau Public or a Streamlit web application.